

Ex.no:4	EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION

Aim:

To perform data inspection, conditional filtering, sorting, statistical summary, group aggregation, correlation analysis, and data visualization on an **air quality dataset** using Pandas and Matplotlib.

Algorithm:

1. Start.
2. Import the required Python libraries: Pandas and Matplotlib.pyplot.
3. Load the dataset city_day.csv into a Pandas DataFrame using pd.read_csv().
4. Inspect the basic structure, initial records, shape, column data types, and numerical summaries using df.head(), df.shape, df.dtypes, and df.describe().
5. Check the missing values present in the air quality dataset.
6. Filter rows based on specific pollution conditions, such as PM2.5 greater than 100 and PM10 greater than 100.
7. Sort the dataset based on PM2.5 in descending order using df.sort_values().
8. Compute statistical metrics for PM2.5 including mean, median, mode, and standard deviation.
9. Group the data by city and calculate the average PM2.5 and PM10 values using df.groupby().
10. Calculate the pairwise correlation matrix for selected numerical air pollutant features.
11. Generate a bar plot for average PM2.5 by city and a scatter plot of PM2.5 versus PM10 using Matplotlib.
12. Display and interpret the obtained results.
13. Stop.

Program:

```
import pandas as pd
import matplotlib.pyplot as plt

# Import Dataset
df = pd.read_csv("/content/city_day.csv")

# Dataset Inspection
print("===== FIRST FIVE RECORDS =====")
display(df.head())

print("\n===== SHAPE OF DATASET =====")
```

```
print(df.shape)

print("\n===== DATA TYPES =====")

print(df.dtypes)

print("\n===== SUMMARY =====")

display(df.describe())

# Missing Values

print("\n===== MISSING VALUES =====")

print(df.isnull().sum())

# Filtering Data

# Records with PM2.5 greater than 100

print("\n===== RECORDS WITH PM2.5 > 100 =====")

high_pm25 = df[df["PM2.5"] > 100]

display(high_pm25.head())

# Records with PM10 greater than 100

print("\n===== RECORDS WITH PM10 > 100 =====")

high_pm10 = df[df["PM10"] > 100]

display(high_pm10.head())

# Sorting Data

print("\n===== SORT BY PM2.5 (DESCENDING) =====")

sorted_df = df.sort_values(by="PM2.5", ascending=False)

display(sorted_df.head())

# Statistical Analysis

print("\n===== PM2.5 STATISTICS =====")

print("Mean :", df["PM2.5"].mean())

print("Median :", df["PM2.5"].median())

print("Mode :")

print(df["PM2.5"].mode())

print("Standard Deviation :", df["PM2.5"].std())

print("\n===== DESCRIPTIVE STATISTICS =====")

display(df.describe())
```

```
# Grouping and Aggregation

print("\n===== AVERAGE PM2.5 BY CITY =====")

city_avg_pm25 = df.groupby("City")["PM2.5"].mean()

display(city_avg_pm25)

print("\n===== AVERAGE PM10 BY CITY =====")

city_avg_pm10 = df.groupby("City")["PM10"].mean()

display(city_avg_pm10)

# Correlation Analysis

print("\n===== CORRELATION =====")

corr = df[

    ["PM2.5", "PM10", "NO", "NO2", "NOx", "CO", "SO2", "O3",

     "Benzene", "Toluene", "Xylene"]

].corr()

display(corr)

# Visualization 1

city_avg_pm25.plot(kind="bar", figsize=(10,5))

plt.title("Average PM2.5 by City")

plt.xlabel("City")

plt.ylabel("Average PM2.5")

plt.xticks(rotation=45)

plt.show()

# Visualization 2

plt.figure(figsize=(8,5))

plt.scatter(df["PM2.5"], df["PM10"])

plt.title("PM2.5 vs PM10")

plt.xlabel("PM2.5")

plt.ylabel("PM10")

plt.show()
```

Output:

```
===== FIRST FIVE RECORDS =====
City      Date      PM2.5  PM10   NO    NO2    NOx   NH3    CO    SO2    O3   Benzene  Toluene  Xylene  AQI  AQI_Bucket
0  Ahmedabad  2015-01-01  NaN    NaN    0.92  18.22  17.15  NaN    0.92  27.64  133.36  0.00    0.02    0.00  NaN    NaN
1  Ahmedabad  2015-01-02  NaN    NaN    0.97  15.69  16.46  NaN    0.97  24.55  34.06   3.68    5.50    3.77  NaN    NaN
2  Ahmedabad  2015-01-03  NaN    NaN    17.40  19.30  29.70  NaN    17.40  29.07  30.70   6.80    16.40   2.25  NaN    NaN
3  Ahmedabad  2015-01-04  NaN    NaN    1.70   18.48  17.97  NaN    1.70   18.59  36.08   4.43    10.14   1.00  NaN    NaN
4  Ahmedabad  2015-01-05  NaN    NaN    22.10  21.42  37.76  NaN    22.10  39.33  39.31   7.01    18.89   2.78  NaN    NaN

===== SHAPE OF DATASET =====
(29531, 16)

===== DATA TYPES =====
City      object
Date      object
PM2.5     float64
PM10      float64
NO         float64
NO2        float64
NOx        float64
NH3        float64
CO         float64
SO2        float64
O3         float64
Benzene    float64
Toluene    float64
Xylene     float64
AQI        float64
AQI_Bucket object
dtype: object

===== SUMMARY =====
PM2.5      PM10      NO      NO2      NOx      NH3      CO      SO2      O3      Benzene  Toluene  Xylene  AQI
count  24933.000000  18391.000000  25949.000000  25946.000000  25346.000000  19203.000000  27472.000000  25677.000000  25509.000000  23908.000000  21490.000000  11422.000000  24850.000000
mean    67.450578    118.127103    17.574730    28.560659    32.309123    23.483476    2.248598    14.531977    34.491430    3.280840    8.700972    3.070128    166.463581
std     64.661449     90.605110    22.785846    24.474746    31.646011    25.684275     6.962884    18.133775    21.694928    15.811136    19.969164    6.323247    140.696585
min      0.040000      0.010000    0.020000    0.010000    0.000000    0.010000    0.000000    0.010000    0.010000    0.000000    0.000000    0.000000    13.000000
25%     28.820000    56.255000    5.630000    11.750000    12.820000    8.580000    0.510000    5.670000    18.860000    0.120000    0.600000    0.140000    81.000000
50%     48.570000     95.680000    9.890000    21.690000    23.520000    15.850000    0.890000    9.160000    30.840000    1.070000    2.970000    0.980000    118.000000
75%     80.590000    149.745000    19.950000    37.620000    40.127500    30.020000    1.450000    15.220000    45.570000    3.080000    9.150000    3.350000    208.000000
max     949.990000   1000.000000   390.680000   362.210000   467.630000   352.890000   175.810000  193.860000   257.730000   455.030000   454.850000   170.370000   2049.000000

===== MISSING VALUES =====
City      0
Date      0
PM2.5     4598

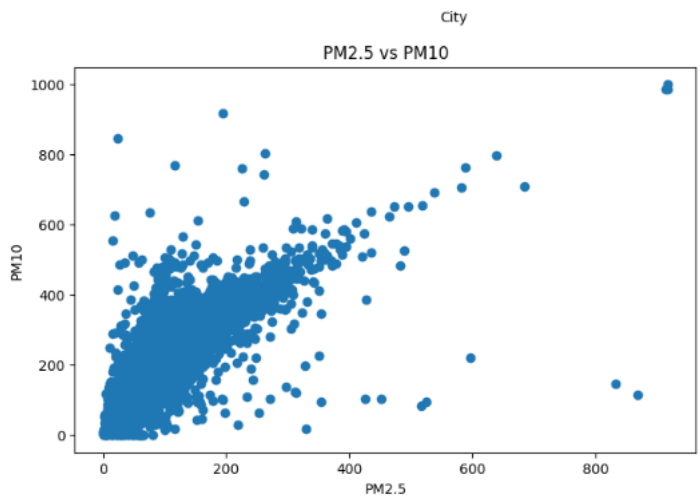
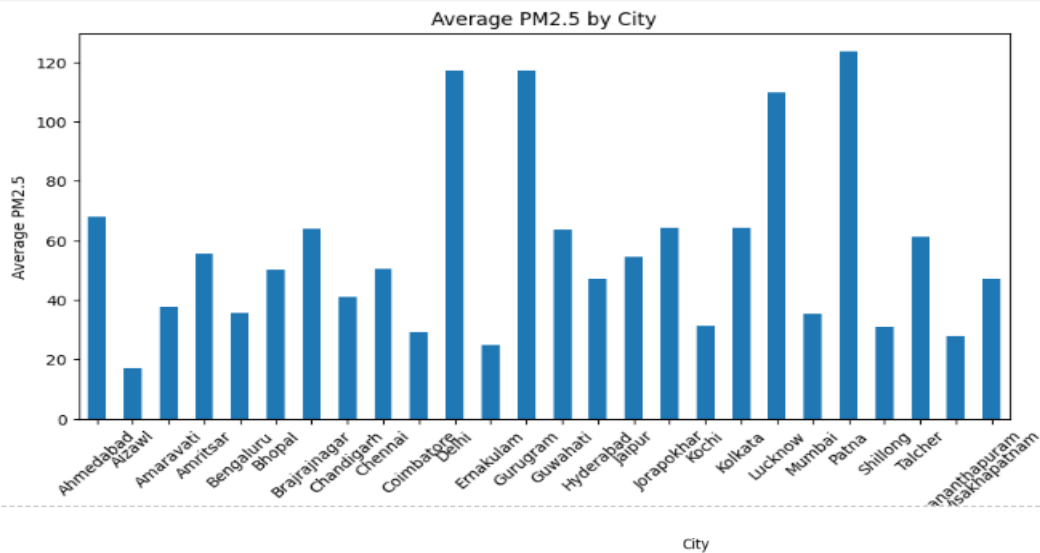
===== RECORDS WITH PM2.5 > 100 =====
City      Date      PM2.5  PM10   NO    NO2    NOx   NH3    CO    SO2    O3   Benzene  Toluene  Xylene  AQI  AQI_Bucket
31 Ahmedabad  2015-02-01  135.99  NaN    43.48  42.08  84.57  NaN    43.48  75.23  102.70  0.40    0.04    25.87  782.0  Severe
32 Ahmedabad  2015-02-02  178.33  NaN    54.56  35.31  72.80  NaN    54.56  55.04  107.38  0.46    0.06    35.61  914.0  Severe
33 Ahmedabad  2015-02-03  139.70  NaN    30.61  28.40  56.73  NaN    30.61  33.79  73.60   0.17    0.03    11.87  660.0  Severe
39 Ahmedabad  2015-02-09  113.93  NaN    4.32   24.27  26.86  NaN    4.32   48.73  39.94   0.02    0.00    3.55   341.0  Very Poor
40 Ahmedabad  2015-02-10  105.39  NaN    1.41   18.21  18.75  NaN    1.41   35.91  56.15   0.00    0.00    3.31   256.0  Poor

===== RECORDS WITH PM10 > 100 =====
City      Date      PM2.5  PM10   NO    NO2    NOx   NH3    CO    SO2    O3   Benzene  Toluene  Xylene  AQI  AQI_Bucket
1594 Ahmedabad  2019-05-14  46.04  141.54  17.26  130.23  49.54  NaN    17.26  107.28  48.32   8.89    52.60    8.78  NaN    NaN
1595 Ahmedabad  2019-05-15  37.55  122.41  15.08  85.12  58.72  NaN    15.08  163.01  48.23   16.44    85.54    14.19  281.0  Poor
1596 Ahmedabad  2019-05-16  33.97  116.32  14.67  79.71  55.61  NaN    14.67  91.26  51.86   15.55    83.89    13.15  330.0  Very Poor
1597 Ahmedabad  2019-05-17  35.48  130.07  18.02  77.61  58.41  NaN    18.02  98.35  38.99   15.88    83.83    13.34  356.0  Very Poor
1598 Ahmedabad  2019-05-18  34.11  138.31  13.27  75.23  51.83  NaN    13.27  88.66  42.22   15.93    82.73    13.44  359.0  Very Poor

===== SORT BY PM2.5 (DESCENDING) =====
City      Date      PM2.5  PM10   NO    NO2    NOx   NH3    CO    SO2    O3   Benzene  Toluene  Xylene  AQI  AQI_Bucket
13041 Gurugram  2017-08-29  949.99  NaN    26.31  28.34  NaN    NaN    0.92  7.44  29.91   0.00    0.00    NaN    NaN    NaN
25729 Shillong  2019-09-03  917.77  985.0  1.26  3.56  1.40  3.64  0.19  5.55  25.20   0.13    0.92    NaN    NaN    NaN
14156 Guwahati  2019-05-04  916.67  1000.0  37.70  13.57  69.67  99.42  0.23  48.03  51.30   86.24    NaN    NaN    NaN    NaN
12504 Gurugram  2016-03-10  914.94  NaN    44.34  63.94  42.71  NaN    1.45  0.64  0.80    NaN    NaN    NaN    676.0  Severe
25728 Shillong  2019-09-02  914.64  985.0  1.32  3.61  1.36  3.63  0.23  5.56  27.96   0.14    0.99    NaN    NaN    NaN

===== PM2.5 STATISTICS =====
Mean : 67.45057794890306
Median : 48.57
Mode :
0 11.0
Name: PM2.5, dtype: float64
Standard Deviation : 64.66144945715158
```

	PM2.5	PM10	NO	NO2	NOx	CO	SO2	O3	Benzene	Toluene	Xylene
PM2.5	1.000000	0.846498	0.433491	0.350709	0.436792	0.089912	0.132325	0.161238	0.023911	0.117080	0.114579
PM10	0.846498	1.000000	0.502349	0.464380	0.527768	0.112588	0.256974	0.244919	0.022265	0.169335	0.081700
NO	0.433491	0.502349	1.000000	0.478070	0.794890	0.212607	0.170322	0.014580	0.035771	0.150857	0.094237
NO2	0.350709	0.464380	0.478070	1.000000	0.627627	0.356521	0.392233	0.293349	0.025260	0.273926	0.171701
NOx	0.436792	0.527768	0.794890	0.627627	1.000000	0.226992	0.238397	0.093170	0.039121	0.189386	0.087398
CO	0.089912	0.112588	0.212607	0.356521	0.226992	1.000000	0.489697	0.041736	0.061861	0.277904	0.154889
SO2	0.132325	0.256974	0.170322	0.392233	0.238397	0.489697	1.000000	0.162142	0.036110	0.296139	0.251195
O3	0.161238	0.244919	0.014580	0.293349	0.093170	0.041736	0.162142	1.000000	0.020255	0.130209	0.111410
Benzene	0.023911	0.022265	0.035771	0.025260	0.039121	0.061861	0.036110	0.020255	1.000000	0.739286	0.415427
Toluene	0.117080	0.169335	0.150857	0.273926	0.189386	0.277904	0.296139	0.130209	0.739286	1.000000	0.421432
Xylene	0.114579	0.081700	0.094237	0.171701	0.087398	0.154889	0.251195	0.111410	0.415427	0.421432	1.000000



Result: